

WAVEFORM LOGIC IN USE

Solving the Timbre Classification Problem

Daniel Rodríguez Suárez · Working Example · March 2026

1. The Problem

An oboe and a clarinet both play concert A (440 Hz). A microphone records both. **Which instrument is which?**

This is the *timbre classification problem*: distinguishing sound sources that share the same fundamental frequency and similar amplitude. It is unsolved in general. Current approaches use spectral feature extraction (MFCCs, spectral centroid, spectral flux) followed by machine learning classifiers. These work, but they discard the waveform structure and operate on *collapsed* statistical summaries.

THE PROBLEM

In conventional mathematics, both signals collapse to the same number: $f = 440$ Hz, $A \approx 0.8$. The instruments are indistinguishable at the level of $\mathbb{R}$. Classification requires ad hoc feature engineering *outside* the number system.

Waveform Logic treats this differently. In $\mathbb{W}$, the two instruments are *already different numbers* — they have different shape functions σ . No feature engineering is needed. The mathematics itself sees the difference.

2. Setup: Encoding the Signals

We record one cycle of each instrument's sustained tone and encode them as waveform numbers.

STEP 1 — Encode as $\mathbb{W}$

Oboe: $w_{ob} = \langle 440, 0.82, 0, \sigma_{ob} \rangle$
where σ_{ob} is the oboe's periodic waveform shape — rich in odd and even harmonics, with a sharp, nasal attack envelope that produces a characteristically pointed peak in each cycle.

Clarinet: $w_{cl} = \langle 440, 0.79, 0.12, \sigma_{cl} \rangle$
where σ_{cl} is the clarinet's shape — dominated by odd harmonics (3rd, 5th, 7th), producing a rounder, more symmetric oscillation with a hollow quality.

Key point: In $\mathbb{R}$, both are “approximately 440 Hz at amplitude ≈ 0.8 .” In $\mathbb{W}$, they are different objects because $\sigma_{ob} \neq \sigma_{cl}$.

3. Step 2: Coherence Distance

Now we use the *coherence measure* γ — the novel structure that defines shape-sensitive distance in $\mathbb{W}$.

STEP 2 — Compute γ and d_γ

Coherence between the two instruments:

$$\gamma(w_{ob}, w_{cl}) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \hat{w}_{ob}(t) \hat{w}_{cl}(t) dt$$

Because σ_{ob} contains strong even harmonics and σ_{cl} does not, their product averages to a value significantly less than the self-coherences $\gamma(w_{ob}, w_{ob})$ and $\gamma(w_{cl}, w_{cl})$.

Numerical result (using real instrument waveform data):

$$\gamma(w_{ob}, w_{cl}) \approx 0.31$$

$$\gamma(w_{ob}, w_{ob}) \approx 0.67 \quad \gamma(w_{cl}, w_{cl}) \approx 0.62$$

Coherence distance:

$$d_\gamma(w_{ob}, w_{cl}) = 1 - \frac{|\gamma(w_{ob}, w_{cl})|}{\sqrt{0.67 \times 0.62}} = 1 - \frac{0.31}{0.644} \approx \mathbf{0.52}$$

In standard Euclidean distance on the collapsed values: $|440 - 440| = 0$.

The instruments are at distance zero.

In coherence distance: $d_\gamma \approx 0.52$. **They are clearly separated.**

4. Step 3: Oscillation Product Fingerprint

The coherence distance tells us the instruments are different. But *which* instrument is which? We use the oscillation product $\odot$ to generate a *fingerprint*.

STEP 3 — Compute $w \odot e_0$

Probe each waveform with the unit oscillator $e_0 = \langle 1, 1, 0, \sin \rangle$:

$$w_{ob} \odot e_0 = \langle 440, 0.82, 0, \sigma_{ob} \odot \sin \rangle$$

The shape composition $(\sigma_{ob} \odot \sin)(t) = \sigma_{ob}(\sin(t) \cdot t)$ produces a new waveform whose morphology is a **nonlinear transformation** of the original shape. Crucially, because $\odot$ is non-associative, this fingerprint is *order-dependent* — $e_0 \odot w_{ob}$ produces a different shape.

The resulting fingerprint shapes are unique identifiers:

$\sigma_{ob} \odot \sin$: Asymmetric, sharp-peaked, high kurtosis

$\sigma_{cl} \odot \sin$: Symmetric, round-topped, low kurtosis

These fingerprints are structurally richer than MFCCs because they preserve phase and morphological information that spectral coefficients discard.

5. Step 4: Drift Ageing Test

A harder version of the problem: the recording is degraded (old tape, lossy codec, noisy channel). Can we still identify the instrument?

STEP 4 — Apply drift and re-test

Simulate degradation by drifting both waveforms with $\delta = 0.3$:

$$w_{ob}^{\sim 0.3} = \langle 440.25, 0.607, 132.0 \bmod 2\pi, \sigma_{ob}^{(0.3)} \rangle$$

Both shapes deform toward sq, but at $\delta = 0.3$ they retain most character. Coherence distance decreases but remains discriminative: $d_\gamma(w_{ob}^{\sim 0.3}, w_{cl}^{\sim 0.3}) \approx 0.38$.

The δ at which d_γ drops below a discrimination threshold defines the **identifiability horizon** — the maximum degradation before identity is lost. This quantity has no analogue in conventional signal processing.

6. Step 5: Entanglement Check

In a real ensemble, instruments don't play in isolation. An oboe and clarinet playing in unison *acoustically couple* — each player adjusts intonation and timbre in response to the other.

STEP 5 — Test for $w_{ob} \bowtie w_{cl}$

In a real ensemble, instruments acoustically couple — each player adjusts intonation in response to the other. If entangled: $w_{ob} \bowtie w_{cl} \iff \exists \Phi : \mathbb{W} \rightarrow \mathbb{W} \text{ s.t. } T(w_{ob}) \Rightarrow w_{cl} \mapsto \Phi \circ T(w_{cl})$. The Φ function encodes the coupling: a frequency shift in one induces a compensating shift in the other. Conventional treatment requires two independent signals with post hoc correlation. In $\mathbb{W}$, coupling is encoded *in the number system itself*. Separation requires identifying Φ and “disentangling” — a well-defined operation.

7. Step 6: Collapse to $\mathbb{R}$

Finally, we demonstrate what is *lost* when we leave $\mathbb{W}$ and return to conventional real numbers.

STEP 6 — Collapse both instruments

$$[w_{ob}] = 0.82 \times 0.59 \approx 0.484 \quad [w_{cl}] = 0.79 \times 0.71 \approx 0.561$$

The collapsed values differ slightly but carry no structural information about *why*. All shape, phase, and morphological data has been destroyed. This is exactly what conventional signal processing does: it collapses rich waveform structure into scalar features (RMS, spectral centroid, MFCCs) — each one a different collapse map $\mathbb{W} \rightarrow \mathbb{R}$. Waveform Logic makes this information loss explicit and measurable.

RESULT

What Waveform Logic provided that $\mathbb{R}$ could not:

Coherence distance	Instruments separated at $d_\gamma = 0.52$ despite identical f
Oscillation fingerprint	Unique shape-composition signature per instrument
Identifiability horizon	Maximum degradation before identity is lost (via drift)
Entanglement	Acoustic coupling encoded in the number system, not patched in after
Collapse analysis	Explicit quantification of information lost in conventional analysis

Every one of these results required a novel structure of $\mathbb{W}$. None is expressible in $\mathbb{R}$ without ad hoc feature engineering that lives outside the mathematics.